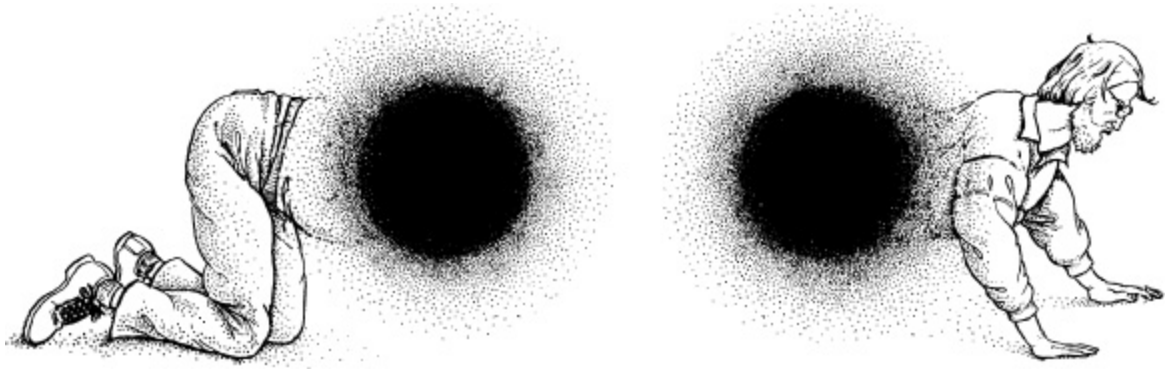
last, had come. In isolation, I let the gnawing emerge from my subconscious and began to examine it: "*How does time decide how to hook itself up through a wormhole?*" That was the nub of the gnaw.

To make this question more concrete, I thought about an example: Suppose that I have a very short wormhole, one whose tunnel through hyperspace is only 30 centimeters long, and suppose that both mouths of the wormhole—two spheres, each 2 meters in diameter—are sitting in my Pasadena living room. And suppose that I climb through the wormhole, head first. From my viewpoint, I must emerge from the second mouth immediately after I enter the first, with no delay at all; in fact, my head is coming out of the second mouth while my feet are still entering the first. Does this mean that my wife, Carolee, sitting there on the living room sofa, will also see my head emerging from the second mouth while my feet are still climbing into the first, as in Figure 14.5? If so, then time "hooks up *through* the wormhole" in the same manner as it hooks up *outside* the wormhole.



14.5 A picture of me crawling through a hypothetical, very short wormhole.

On the other hand, I asked myself, isn't it possible that, although the trip through the wormhole takes almost no time as seen by me, Carolee must wait an hour before she sees me emerge from the second mouth; and isn't it also possible that she sees me emerge an hour before I entered? If so, then time would be hooked up *through* the wormhole in a different manner than it hooks up *outside* the wormhole.

What could possibly make time behave so weirdly? I asked myself. On the other hand, why shouldn't it behave in this way? Only the laws of physics know the answer, I reasoned. Somehow, I ought to be able to deduce from the laws of physics just how time will behave.